

Zepto Inventory, Pricing & Revenue Analysis Using SQL

Project Overview

This project analyzes Zepto's product inventory dataset using MySQL. The objective is to extract meaningful business insights related to product categories, stock availability, discounts, pricing, inventory weight, and revenue estimation.

The project demonstrates the use of SQL for data cleaning, data exploration, aggregation, filtering, and business analytics.

Objectives

- Analyze product categories available on Zepto.
 - Identify in-stock and out-of-stock products.
 - Detect duplicate product names.
 - Clean and validate pricing data.
 - Find products with the highest discounts.
 - Analyze high-value products that are out of stock.
 - Estimate category-wise revenue.
 - Calculate average discounts across categories.
 - Determine price per gram for products.
 - Categorize products based on weight.
 - Calculate total inventory weight by category.
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Dataset Information

Table Name

zepto1

Columns

Column Name	Description
sku_id	Unique product identifier
category	Product category
name	Product name

Column Name	Description
mrp	Maximum Retail Price
discountPercent	Discount percentage
availableQuantity	Available stock quantity
discountedSellingPrice	Selling price after discount
weightInGms	Product weight in grams
outOfStock	Product stock status
quantity	Quantity per package

Technologies Used

- MySQL
- SQL Queries
- Data Analysis
- Data Cleaning
- Aggregation Functions

Key SQL Concepts Used

- SELECT
- DISTINCT
- WHERE
- ORDER BY
- GROUP BY
- HAVING
- Aggregate Functions (COUNT, SUM, AVG)
- CASE Statements
- UPDATE Statements
- Data Cleaning Techniques

Business Questions Solved

1. Total Number of Products

Count the total records available in the dataset.

2. Product Categories Analysis

Identify all unique product categories.

3. Inventory Status Analysis

Compare products that are in stock versus out of stock.

4. Duplicate Product Detection

Find products appearing multiple times with different SKUs.

5. Data Quality Check

Detect products having zero MRP or selling price.

6. Highest Discount Products

Identify top products offering maximum discounts.

7. Out-of-Stock Premium Products

Find expensive products currently unavailable.

8. Revenue Estimation

Estimate total potential revenue generated by each category.

9. Discount Analysis

Determine categories offering the highest average discounts.

10. Price Per Gram Analysis

Identify products providing the best value for money.

11. Weight-Based Classification

Classify products into:

- Low Weight
- Medium Weight
- Bulk Weight

12. Inventory Weight Analysis

Calculate total inventory weight available in each category.

Sample Insights

- Categories with higher discounts attract more customers.
 - Out-of-stock premium products indicate potential demand.
 - Revenue estimation helps prioritize high-performing categories.
 - Price-per-gram analysis assists in value comparison.
 - Inventory weight analysis supports warehouse planning.
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Project Outcomes

This project demonstrates how SQL can be used to:

- Perform exploratory data analysis.
 - Clean and validate datasets.
 - Generate business insights.
 - Support inventory and pricing decisions.
 - Estimate revenue opportunities.
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Future Enhancements

- Build an interactive dashboard using Power BI.
 - Create sales forecasting models.
 - Integrate Python for advanced analytics.
 - Perform customer purchase behavior analysis.
 - Develop category-wise performance dashboards.
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Author

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Skills:

- SQL
 - MySQL
 - Power BI
 - Excel
 - Data Visualization
 - Data Analysis
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This project was created for learning and demonstrating SQL-based retail data analytics using a Zepto product dataset.